

Adaptation, Insurer Solvency, and Affordability

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Abstract

I analyze the growing tension between insurer solvency and insurance affordability in property markets exposed to climate change. I develop a structural model where risk-averse households choose between purchasing insurance and investing in physical risk adaptation, while competitive insurers face regulatory capital requirements for systemic climate risk. My model demonstrates two findings: (1) high frictional cost of regulatory capital creates a high entry price for insurance that is relatively insensitive to individual adaptation, and (2) below a specific wealth threshold, both physical adaptation and insurance coverage collapse as households lack the liquidity to clear the solvency-loaded premium. Through policy counterfactuals, I find that premium subsidies are counterproductive; by making insurance artificially cheap, they cause households to increase financial coverage while decreasing physical adaptation effort. In contrast, adaptation grants that directly target the liquidity friction are highly effective, allowing households to jump to a higher state of resilience, which improves both household welfare and market-wide financial stability.

Keywords: Climate Finance, Property Insurance, Adaptation, Solvency, Affordability, Moral Hazard.

JEL Classification: G22, G28, Q54.

1 Introduction

Climate change is increasingly straining property insurance markets, creating a dual crisis of solvency and affordability. In high-risk regions like Florida and California, major insurers are withdrawing coverage, citing the unmanageable cost of capital required to back policies against correlated catastrophes like hurricanes and wildfires. For households, this translates into a double jeopardy: they face escalating physical threats while the skyrocketing cost of the regulatory capital required to insure those threats renders financial protection unaffordable. This paper investigates the critical link between the regulatory pressures on insurers and the financial constraints on households, arguing that their interaction creates a structural barrier to private climate change adaptation.

This real-world tension motivates my central research questions. First, can insurer solvency regulations, designed to ensure financial stability, distort the pricing of insurance and mute the incentives for households to invest in risk-reducing adaptation? Second, under what conditions do these market forces create an Affordability Trap, a low-wealth equilibrium where households are forced to choose between physical resilience and financial survival? Finally, what is the most effective form of government intervention? As climate risk becomes systemic, understanding how policy should incentivize both adaptation and affordable risk-transfer, promoting a more stable and resilient residential property market, is the core focus of this research.

To answer these questions, I develop and numerically solve a structural model of the property insurance market. My model features risk-averse households with Constant Absolute Risk Aversion (CARA) preferences who choose insurance coverage and an adaptation level subject to a strict budget constraint. They purchase coverage from competitive insurers who must satisfy a Value-at-Risk (VaR) solvency constraint by holding costly capital. The primary methodological contribution lies in endogenizing the feedback loop between individual adaptation and the unit premium. The price a household pays depends on the insurer's total solvency cost, which is determined by the aggregate standard deviation of the market. Because this cost is socialized across all policyholders, it creates a Solvency Friction that decouples individual premiums from individual mitigation efforts. This framework allows me to bridge the corporate finance of insurance ([Froot, 2007]; [Cummins & Weiss, 2009]), household finance ([Guiso & Sodini, 2013]), and climate economics ([Kousky, 2014]; [Heal & Millner, 2014]).

My findings reveal an interaction between these frictions. First, I identify an Affordability Trap threshold. Below this wealth level, the entry price of insurance—driven by high solvency loadings—exhausts the household's budget. Consequently, both adaptation and insurance coverage decline steeply, leaving the most vulnerable households exposed to both physical ruin and financial insolvency. Second, the model highlights that even unconstrained households choose rational under-insurance ($q^* < 1$) because the solvency

loading makes insurance actuarially unfair. Most importantly, counterfactual simulations demonstrate that traditional premium subsidies are counterproductive. In my model, a 20% premium subsidy induces moral hazard, causing households to increase their financial coverage to 100% while decreasing their physical adaptation effort. In contrast, adaptation grants that directly resolve the liquidity friction allow households to jump to a higher level of resilience. This structural improvement lowers the systemic risk of the entire pool and establish a sustainable insurance market.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 details the theoretical model. Section 4 presents the simulation, parameterization, and results. Section 5 discusses an empirical strategy for testing the model’s predictions. Section 6 concludes.

2 Literature Review

My work is situated at the intersection of three areas.

First, I build on the extensive literature on insurance economics and risk management, particularly the effects of capital requirements. Scholars have long studied the role of capital in insurance markets, especially for catastrophic risk ([Froot, 2007]). Regulatory frameworks like Solvency II and NAIC risk-based capital explicitly link capital requirements to tail risk (VaR or TVaR measures). Prior research has established a clear link between capital adequacy and market outcomes. For instance, Eastman and Kim (2023) [Eastman & Kim, 2023] find that stringent capital requirements lead to increased insurance premiums. Similarly, Gron (1994) [Gron, 1994] shows that an unexpected decrease in financial capacity results in higher prices and profitability. The literature has explored these dynamics across various regulatory landscapes, with Barbu (2023) examining the transition from Solvency I to Solvency II in the European Union, and Tang (2022) [Tang, 2022] investigating competitive dynamics among U.S. state jurisdictions. Research has also delved into how capital regulations influence specific investment and operational decisions ([Becker et al., 2022]; [Niehaus, 2018]). My previous work (Wu, 2026) [Wu, 2026] further disentangles the roles of capital regulation and credit ratings in mitigating insolvency risk, finding that tightening capital requirements improves solvency but raises prices, and that heightened rating salience alone cannot fully substitute for regulatory stability. While this body of literature establishes the causal impact of capital regulation on solvency, pricing, and insurer behavior, a gap remains in understanding how these capital costs, driven by systemic climate perils, are ultimately passed on in premiums and how this pricing signal (or lack thereof) directly impacts household adaptation incentives. My model explicitly addresses this gap by endogenizing the link between capital requirements, systemic climate risk, and household adaptation decisions.

Secondly, I contribute to the household finance literature. A large body of work doc-

uments how liquidity constraints and wealth levels affect household portfolio choices and risk management decisions ([Guiso et al., 2002]; [Guiso & Sodini, 2013]). This literature highlights how financial frictions can lead to suboptimal decisions, particularly for low-wealth households. However, most existing research does not explicitly model the joint decision-making for real asset adaptation and financial risk transfer. I apply this lens to the choice between insurance and adaptation, showing that these decisions are not independent and can be severely distorted by affordability constraints. My paper extends this literature by demonstrating how a solvency friction at the insurer level translates into an affordability trap at the household level, hindering adaptation, which is a novel finding in this context.

Finally, my paper speaks to the growing literature on the economics of climate adaptation. This field examines the costs and benefits of adaptation measures and the role of public policy in incentivizing them ([Heal & Millner, 2014]; [Kousky, 2014]). While much of this literature focuses on public infrastructure or the aggregate economic costs of disasters, I focus on private, household-level adaptation and the critical role of insurance markets as a transmission mechanism for policy. Existing work often assumes that price signals from insurance are sufficient or explores direct subsidies in isolation. My finding that adaptation subsidies strictly dominate premium subsidies provides a clear micro-founded rationale for programs like FEMA’s Hazard Mitigation Grant Program, as it directly addresses the identified liquidity constraint without exacerbating moral hazard, a crucial distinction that has not been fully explored through an integrated structural framework linking insurer solvency and household behavior.

3 A Model of Insurance and Adaptation

I develop a static structural model of a property insurance market (e.g., a coastal region exposed to hurricanes) populated by N ex-ante identical households and a representative, competitive insurance sector.

3.1 Risk, Adaptation, and Geographical Dependency

Each household i faces an individual property loss $L_i(a)$ which is assumed to be Normally distributed, $L_i(a) \sim \mathcal{N}(\mu(a), \sigma^2)$. The household can choose a continuous adaptation level $a \geq 0$ at a convex cost $c(a) = \frac{1}{2}ka^2$. Adaptation directly reduces the expected property loss:

$$\mu(a) = \mu_0 - \theta_1 a \tag{1}$$

where μ_0 is the baseline expected loss and $\theta_1 > 0$ measures adaptation efficacy.

The critical friction in catastrophe insurance is geographical dependency. If a hurricane strikes, losses are highly correlated. Let the correlation coefficient between any

two households i and j be $\rho(a)$. I assume physical adaptation reduces this systemic risk: $\rho'(a) < 0$. For instance, an entire neighborhood of unadapted homes will flood simultaneously during a storm surge, but if homes are elevated, only those hit by localized, idiosyncratic wave action will suffer severe damage.

3.2 Insurer's Problem: Solvency and Pricing

A representative competitive insurer underwrites policies for the entire market. The aggregate loss is $\mathcal{L} = \sum_{i=1}^N L_i$. The variance of this aggregate loss is derived from the sum of variances and all pairwise covariances:

$$\text{Var}(\mathcal{L}) = \sum_{i=1}^N \text{Var}(L_i) + \sum_{i \neq j} \text{Cov}(L_i, L_j) = N\sigma^2 + N(N-1)\rho(a)\sigma^2 \equiv \sigma_{agg}^2(a) \quad (2)$$

where the number of distinct pairs (i, j) with $i \neq j$ is $N(N-1)$.

In a competitive market, the insurer sets premiums such that expected profits are zero. The insurer's total revenue must cover total expected claims plus the frictional cost of holding regulatory capital. Let's assume every identical household chooses the same coverage level q . The total underwritten portfolio for the insurer is therefore $q\mathcal{L}$.

Regulatory solvency requires the insurer to hold capital K against unexpected losses. This is modeled as a Value-at-Risk (VaR) constraint, where capital must be sufficient to cover losses up to the α -percentile of the distribution of the underwritten portfolio, net of expected claims.

$$K = \text{VaR}_\alpha(q\mathcal{L}) - \mathbb{E}[q\mathcal{L}] \quad (3)$$

Since the individual losses L_i are Normal, the aggregate underwritten loss $q\mathcal{L}$ is also Normal, with mean $\mathbb{E}[q\mathcal{L}] = qN\mu(a)$ and variance $\text{Var}(q\mathcal{L}) = q^2\sigma_{agg}^2(a)$. The standard deviation is $q\sigma_{agg}(a)$. The VaR is the mean of the distribution plus Z_α standard deviations:

$$\text{VaR}_\alpha(q\mathcal{L}) = \mathbb{E}[q\mathcal{L}] + Z_\alpha \cdot \sqrt{\text{Var}(q\mathcal{L})} = qN\mu(a) + Z_\alpha q\sigma_{agg}(a) \quad (4)$$

Substituting this into the definition of capital gives:

$$K = (qN\mu(a) + Z_\alpha q\sigma_{agg}(a)) - qN\mu(a) = qZ_\alpha\sigma_{agg}(a) \quad (5)$$

Holding this capital is frictional and incurs a proportional cost $\tau > 0$ (e.g., due to taxes or agency costs). The total cost of capital for the insurer is τK . The zero-profit condition for the entire market is:

$$\text{Total Premium Income} = \text{Total Expected Claims} + \text{Cost of Capital} \quad (6)$$

Let $p(a)$ be the *unit price of insurance* (the price for full coverage, $q = 1$). A household purchasing coverage q pays a total premium of $q \cdot p(a)$. The insurer's total premium income from all N households is $N \cdot q \cdot p(a)$. Substituting all terms into the zero-profit condition:

$$N \cdot q \cdot p(a) = qN\mu(a) + \tau (qZ_\alpha \sigma_{agg}(a)) \quad (7)$$

We can divide both sides by $N \cdot q$ (assuming $q > 0$) to solve for the unit price $p(a)$:

$$p(a) = \mu(a) + \frac{\tau Z_\alpha}{N} \sigma_{agg}(a) \quad (8)$$

Defining the shadow price of the solvency constraint as $\lambda = \tau Z_\alpha$, we arrive at the final expression for the equilibrium unit premium:

$$p(a) = \mu(a) + \frac{\lambda}{N} \sqrt{N\sigma^2 + N(N-1)\rho(a)\sigma^2} \quad (9)$$

The first term, $\mu(a)$, is the standard actuarially fair component. The second term is the solvency loading: it is the per-household, per-unit-of-coverage allocation of the total frictional cost of regulatory capital. This premium structure explicitly links a household's unit cost of insurance to the systemic risk $\rho(a)$ that collective adaptation mitigates.

3.3 Risk-Averse Household's Problem

The representative household has an initial wealth Y and exhibits Constant Absolute Risk Aversion (CARA) preferences, $u(W) = -\exp(-\gamma W)$. The household chooses an adaptation level a and an insurance coverage fraction q to maximize expected utility, subject to a strict liquidity (affordability) constraint.

The household's final wealth is given by:

$$W = Y - \frac{1}{2}ka^2 - qp(a) - (1-q)L_i(a) \quad (10)$$

Because $L_i(a)$ is Normally distributed, final wealth W is also Normal with mean $\mathbb{E}[W] = Y - \frac{1}{2}ka^2 - qp(a) - (1-q)\mu(a)$ and variance $\text{Var}(W) = (1-q)^2\sigma^2$.

A standard property of CARA utility over Normal distributions is that maximizing expected utility $\mathbb{E}[-e^{-\gamma W}]$ is mathematically equivalent to maximizing the Certainty Equivalent (CE), defined as $\mathbb{E}[W] - \frac{\gamma}{2}\text{Var}(W)$. Therefore, the household's optimization problem simplifies to:

$$\max_{a,q} \quad Y - \frac{1}{2}ka^2 - q \cdot p(a) - (1-q)\mu(a) - \frac{\gamma}{2}(1-q)^2\sigma^2 \quad (11)$$

$$\text{s.t.} \quad q \cdot p(a) + \frac{1}{2}ka^2 \leq Y \quad (12)$$

Equation (12) is the affordability constraint, stipulating that the combined upfront cost of the insurance premium and physical adaptation cannot exceed the household’s available liquid wealth.

4 Simulation and Results

I numerically solve the household’s optimization problem to illustrate the model’s key mechanisms and evaluate policy interventions.

4.1 Parameterization

I choose a set of stylized but plausible parameters to represent a high-risk coastal property market, summarized in Table 1. The household exhibits high risk aversion ($\gamma = 1.2$), ensuring they prioritize protecting wealth from catastrophic variance. The cost of physical adaptation is relatively steep ($k = 2.5$). The baseline risk is high ($\mu_0 = 10$) with significant geographical dependency ($\rho_0 = 0.8$). The cost of insurer capital is set to $\tau = 15\%$, reflecting the high cost for catastrophe-exposed business lines.

Table 1: Simulation Parameter Values		
Parameter	Symbol	Value
<i>Household</i>		
Risk Aversion	γ	1.2
Adaptation Cost	k	2.5
<i>Risk & Adaptation</i>		
Baseline Expected Loss	μ_0	10.0
Idiosyncratic Std. Dev.	σ	5.0
Baseline Correlation	ρ_0	0.8
Adaptation Efficacy (Mean)	θ_1	1.0
Adaptation Efficacy (Corr.)	θ_ρ	0.5
<i>Insurer & Market</i>		
Market Size	N	100
Cost of Capital	τ	0.15
VaR Confidence Level	α	99% ($Z_\alpha = 2.33$)

4.2 Counterfactuals and Results

I conduct three counterfactual experiments corresponding to my main research questions. The results are displayed graphically in Figure 1 and numerically in Table 2.

RQ1: The Solvency Friction. Panel A of Figure 1 demonstrates how the cost of regulatory capital creates a solvency friction that is passed on to consumers. The red line (high cost of capital, $\tau = 25\%$) is uniformly above the blue line (low cost,

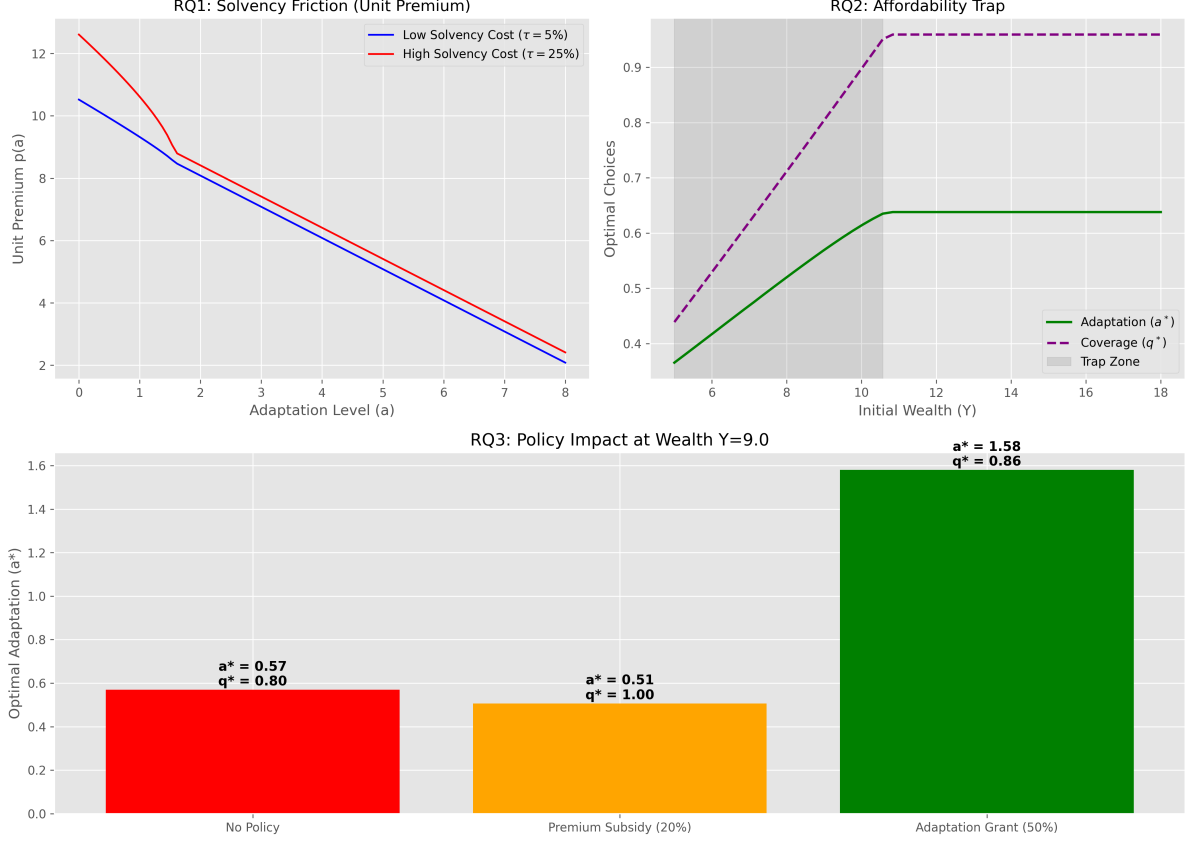


Figure 1: Simulation Results for the Three Research Questions

Notes: This figure displays the results of the three numerical experiments. Panel A shows the relationship between adaptation and the unit insurance premium under a low ($\tau = 5\%$) and high ($\tau = 25\%$) cost of insurer capital. Panel B shows the household's optimal adaptation (a^*) and insurance coverage (q^*) as a function of initial wealth (Y). Panel C compares the optimal adaptation level and coverage for a liquidity-constrained household ($Y = 9.0$) under three policy regimes.

$\tau = 5\%$), confirming that higher capital costs for insurers lead to higher unit premiums for households. This elevated entry price for insurance is a primary driver of the affordability problem. Both curves are downward sloping, showing that adaptation mitigates this by reducing expected losses and systemic risk.

RQ2: The Affordability Trap. Panel B reveals the model's central mechanism. Because physical adaptation is relatively expensive ($k = 2.5$) and the household is highly risk-averse, we observe a sharp structural break at a wealth threshold of $Y \approx 11.5$.

- **Below the threshold ($Y < 11.5$, Trap Zone):** Households are strictly liquidity-constrained. The affordability constraint binds tightly, forcing the household to ration its budget between physical adaptation (a^*) and financial insurance (q^*). As wealth decreases within this zone, we observe simultaneous, steep declines in both adaptation and coverage. To avoid the catastrophic utility loss of an uninsured systemic shock, the highly risk-averse household optimally allocates a larger share of its shrinking budget to purchasing insurance, forcing a drastic reduction in physical risk reduction. This illustrates the vicious cycle of the affordability trap: the

household is too poor to invest in making itself safer.

- **Above the threshold ($Y \geq 11.5$):** The household has sufficient wealth to clear the friction. Optimal adaptation reaches an unconstrained interior solution ($a^* \approx 0.64$). Due to Constant Absolute Risk Aversion (CARA) preferences, this choice of risk mitigation remains flat and independent of wealth once it is affordable.

Crucially, the graph shows that optimal insurance coverage (q^*) is consistently less than 1.0 (approximately 0.96 in the unconstrained state). This is mathematically consistent and highlights a secondary market failure. Because the solvency loading forces the premium strictly above the actuarially fair expected loss, risk-averse households rationally choose to retain a fraction of the risk (under-insurance), as the marginal cost of full coverage exceeds its marginal utility benefit.

RQ3: Optimal Policy Mechanism. Panel C and Table 2 powerfully illustrate the differential impact of policies for a household deep in the trap ($Y = 9.0$).

Table 2: Impact of Policies Across Different Wealth Levels.

Wealth	Policy	a^*	q^*	CE
9.00	Baseline	0.57	0.80	-2.43
9.00	20% Premium Sub.	0.51	1.00	0.05
9.00	50% Adapt. Grant	1.58	0.86	-1.49
11.50	Baseline	0.64	0.96	0.44
11.50	20% Premium Sub.	0.51	1.00	2.55
11.50	50% Adapt. Grant	1.58	0.99	1.27
15.00	Baseline	0.64	0.96	3.94
15.00	20% Premium Sub.	0.51	1.00	6.05
15.00	50% Adapt. Grant	1.58	0.99	4.77

Notes: This table presents the optimal adaptation (a^*), insurance coverage (q^*), and Certainty Equivalent (CE) for households at different initial wealth (Y) levels under various policy interventions. The 'Baseline' represents no government intervention. '20% Premium Sub.' indicates a 20% subsidy on the insurance premium. '50% Adapt. Grant' denotes a 50% subsidy on the cost of adaptation. The parameter values used in the simulation are detailed in Table 1.

Discussion of Table and Policy Results. Table 2 provides a detailed numerical analysis of the household's choices and allows for a direct welfare comparison using the Certainty Equivalent (CE).

- **No Policy (Baseline):** At a constrained wealth of $Y = 9.0$, the household is trapped into a suboptimal mix of partial adaptation ($a^* = 0.57$) and significant under-insurance ($q^* = 0.80$), resulting in a highly negative CE (-2.43).
- **The Moral Hazard of Premium Subsidies:** A 20% premium subsidy clearly highlights the dangers of poorly designed intervention. As seen in Panel C and Table 2, this policy causes the household to increase insurance coverage to full protection

($q^* = 1.00$). However, this artificially cheapened premium reduces the financial return on risk mitigation. Consequently, the household exhibits moral hazard and *decreases* its physical adaptation effort from $a^* = 0.57$ down to $a^* = 0.51$. While the household’s CE rises significantly (to 0.05), this is entirely driven by the direct financial wealth transfer of the subsidy, rather than an improvement in underlying systemic risk. This policy effectively discourages resilience.

- **The Efficacy of Adaptation Grants:** A 50% subsidy on the upfront cost of physical adaptation directly targets the binding affordability constraint. This breaks the trap entirely. The household jumps to a dramatically higher level of adaptation ($a^* = 1.58$). Furthermore, because this physical adaptation significantly lowers the underlying systemic risk, the true insurance premium falls, allowing the constrained household to simultaneously purchase higher coverage ($q^* = 0.86$ vs baseline 0.80).

While the direct financial transfer of the premium subsidy yields a higher instantaneous CE for the individual household, the adaptation grant achieves a more profound structural result. It induces a massive increase in physical safety (a^* almost triples) and improves the risk profile of the entire insurance pool, a systemic benefit that pure financial subsidies fail to generate.

5 Potential Empirical Strategy

The theoretical predictions of my model—specifically the existence of a sharp wealth threshold for adaptation and the moral hazard induced by premium subsidies—provide a clear roadmap for empirical testing. I propose a two-pronged empirical approach combining reduced-form causal inference with structural estimation.

First, to identify the Affordability Trap, I suggest searching for non-linearities in adaptation behavior relative to household wealth or income. Using granular property-level data from sources like CoreLogic or Zillow, one could proxy adaptation (a^*) through building permits for retrofitting or structural characteristics (e.g., elevation, roof type). By merging this with Census-tract level wealth data, a regression discontinuity-style design or a threshold regression could test whether adaptation efforts jump discontinuously at a specific wealth level, as predicted by Panel B of Figure 1.

Second, to test the Optimal Policy predictions, the rollout of the NFIP’s Risk Rating 2.0 provides a quasi-natural experiment. This policy shifted many households from subsidized to actuarially-based (and solvency-loaded) premiums. A Difference-in-Differences (DiD) design could compare households that lost premium subsidies against those that did not. My model predicts that those losing the subsidy ($s_p \downarrow$) should, if not wealth-constrained, actually increase their physical adaptation efforts due to the removal of moral hazard—a counterintuitive but testable hypothesis.

Finally, I intend to extend this work through structural estimation using the Simulated Method of Moments (SMM). By matching the model-predicted choices of a^* and q^* across different wealth deciles to observed market data, I can back out the latent structural parameters: the cost coefficient of adaptation (k) and the coefficient of risk aversion (γ). This would allow for precise quantification of the solvency loading (λ) and its role in suppressing market-wide resilience.

6 Conclusion and Discussion

This paper has developed a unified structural framework to analyze the interaction between insurer solvency regulations, household wealth constraints, and climate change adaptation. By endogenizing the link between individual risk-reduction and the aggregate cost of regulatory capital, I have identified two fundamental frictions that threaten the stability of property insurance markets.

My findings demonstrate that the Affordability Gap is not merely a product of high expected losses, but is exacerbated by a Solvency Friction. The high frictional cost of holding capital against systemic, correlated climate risks is passed on to households as a high unit premium. For the most vulnerable households, this creates an Affordability Trap: the high cost of insurance exhausts their liquid wealth, leaving them unable to afford the upfront costs of physical adaptation (a^*). As wealth declines, these households are forced to ration their budgets, leading to a simultaneous collapse in both physical resilience and financial protection (q^*).

The policy implications of this model are stark. I show that traditional interventions focused on making insurance affordable through premium subsidies (s_p) are largely counterproductive. While they provide a short-term wealth transfer, they induce significant moral hazard, causing households to reduce their physical mitigation efforts. In my simulation, a 20% premium subsidy led to a decrease in adaptation from 0.57 to 0.51. In contrast, adaptation grants (s_a) that directly target the upfront cost of resilience are highly effective. By resolving the liquidity friction, these grants allow households to jump to a higher state of resilience ($a^* = 1.58$), which in turn lowers the systemic risk of the insurer's pool and allows for higher coverage levels.

My model is necessarily stylized and could be extended in several ways. A dynamic framework could incorporate the long-term benefits of adaptation and the evolution of climate risk. Incorporating heterogeneous agents would allow for a richer analysis of the distributional consequences of climate change and adaptation policy. Finally, taking the model to the data, as outlined in Section 5, remains a crucial next step to quantify the magnitude of these effects and inform real-world policy design.

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